Machine-Verified Integer Foundations for Qameah Spectral Theory: 1,296 Lean 4 Declarations

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Abstract

We present a Lean 4 / Mathlib formalization library containing 1,215 theorem and lemma declarations, 81 definitions, spanning 76 files, with zero `sorry` placeholders. The library formally verifies the integer arithmetic skeleton of the QHOTS (Qameah Harmonic Oscillator Theory of Symmetry) framework, which derives physical constants from the topology of an E8-based solid-state vacuum. Three proof strategies—called *hammers*—cover all declarations: (1) *integer arithmetic* (`norm_num`, `native_decide`, `omega`), (2) *algebraic number theory* via Mathlib's `goldenRatio` and `linear_combination`, and (3) *transcendental interval analysis* using Mathlib's π-bounds to propagate tight inequalities. Hammer 3 yields the main result of this paper: $\lfloor 6\pi^5 \rfloor = 1836$, a machine-verified certification that the integer shadow of the classical proton–electron mass-ratio formula equals the observed mass ratio. Additional highlights include the first machine-verified proof that $\{S, A\} = 0$ for the 11×11 $\mathrm{GL}_2(\mathbb{Z}_{11})$ Qameah (the Qameah of order N: the $\mathrm{GL}(2, \mathbb{Z}_N)$ magic square with parameter $s = (N+1)/2$; a distinguished member of the de la Loubère family satisfying the circulant criterion)—a structural identity central to Qameah spectral theory—and a Pochhammer generating-function identity (T95, PILLAR #71) relating the elementary symmetric polynomials of the spectral half to products of binomial coefficients. The library is compiled against Lean 4 v4.14.0 and Mathlib v4.14.0.

1 Introduction

The QHOTS framework [3] derives physical constants from E8 Lie algebra invariants, the 11×11 Qameah magic square, and φ (the golden ratio). Empirical comparisons with measured values are outside the scope of formal proof; the purely mathematical relationships—divisibility, factorizations, polynomial identities, floor values of transcendental expressions—are not. This paper formalizes that integer skeleton in Lean 4 [2] / Mathlib [1], producing 1,296 declarations (1,215 theorems and lemmas, 81 definitions, 0 private lemmas) across the following areas:

- E8 and D4 Lie algebra integer data (dimension, Coxeter number, root count, rank–Coxeter pairing uniqueness).
- Arithmetic of $\alpha = 28/3837$: the denominator decompositions $128 \times 30 - 3 = 3837$ and $137 \times 28 + 1 = 3837$, the master polynomial factorization $r^3 - 4r - 480 = (r-8)(r^2+8r+60)$, the Clifford Bridge $S_3(11) = 2048 = 2^{11}$ (where $S_3(n) = (n^2-1)(2n^4+23n^2+191)/1890$ is the third even-power-sum of $\csc^2(k\pi/n)$), and the irreducibility $\gcd(28, 3837) = 1$.
- The integer shadow $\lfloor 6\pi^5 \rfloor = 1836$ of the classical mass-ratio term, established via interval arithmetic.

- The anticommutator identity $\{S, A\} = 0$—the N=11 instance of the general algebraic theorem (proved in the companion paper [3])—independently verified by finite matrix enumeration in Lean 4.
- The Pochhammer generating-function identity T95 for spectral symmetric polynomials of Qameah magic squares of orders $N = 5, 7, 11$.
- Golden ratio power identities $\varphi^n = F_n\,\varphi + F_{n-1}$ for $n = 2, \ldots, 5$, where F_n are Fibonacci numbers.
- 41 tiers of supplementary arithmetic (Tiers 3–43), covering magic square properties, exceptional algebra dimensions, McKay correspondence group orders, Euler totients, and combinatorial identities relating 120, 240, 480 to the E8 root system.

The library is publicly available at `https://github.com/Insightful-Calculations/qhots_v7_calc`; scope exclusions are listed in Section 6.

2 Library Structure

The library consists of 76 Lean 4 source files organized into two layers: a set of 15 *named modules* covering distinct mathematical topics, and 41 *tier files* (Tier3 through Tier43) that accumulate arithmetic identities in a milestone-driven sequence. A master import file (`Registry.lean`) imports all other modules.

2.1 Named modules

2.2 Tier files (Tiers 3–43)

The 41 tier files (4–34 declarations each) accumulate arithmetic identities in a milestone-driven sequence: magic square combinatorics, exceptional Lie algebra dimensions [6], McKay correspondence [5] group orders, pentagon–hexagon identities, and magic constant formulas $M_N = N(N^2 + 1)/2$ for $N = 3, 5, 7, 9, 11, 13$.

2.3 Registry and build

`Registry.lean` is a master import file that imports all 74 other modules. Building it with

```
lake build
```

in Lean 4 v4.14.0 with Mathlib v4.14.0 as the sole dependency compiles all 1,296 declarations with zero errors and zero `sorry` placeholders. The library compiles under `lake build` with Lean 4.14.0 and Mathlib 4.x; no continuous integration is currently configured.

3 Three-Hammer Proof Strategy

All 1,296 declarations are discharged by three complementary *hammers*.

3.1 Hammer 1: Integer Arithmetic

The majority of declarations are pure arithmetic facts about natural numbers or integers. Lean 4's `norm_num` tactic discharges any closed arithmetic goal directly, while `native_decide` handles goals of the form $\forall x \in S,\ P(x)$ when S is a finite type and P is decidable. Compiled native code is generated for the decision procedure, making `native_decide` practical even for products over $|S| = 11^2 = 121$ index pairs.

Example 1. *The Clifford Bridge theorem states that the power sum* $S_3(11) = 2048 = 2^{11}$*, where* $S_3(n) = (n^2 - 1)(2n^4 + 23n^2 + 191)/1890$.

```
def S3 (n : Nat) : Rat :=
  (n^2 - 1) * (2*n^4 + 23*n^2 + 191) / 1890

theorem clifford_bridge : S3 11 = 2048 := by norm_num [S3]

theorem clifford_bridge_unique_11 :
    forall p in ({3,5,7,11,13,17,19,23,29,31,37,41,43,47} : Finset Nat)
        ,
    S3 p = 2^p <-> p = 11 := by native_decide
```

3.2 Hammer 2: Algebraic Number Theory (φ via `goldenRatio`)

For identities involving the golden ratio $\varphi = (1+\sqrt{5})/2$, we use Mathlib's `Data.Real.GoldenRatio` module, which exports φ as `goldenRatio` and the defining equation as `gold_sq`: $\varphi^2 = \varphi + 1$. Higher powers reduce via `linear_combination` with a carefully chosen polynomial multiple of `gold_sq`.

Example 2. *The identity* $\varphi^5 = 5\varphi + 3$ *(Fibonacci coefficients* $F_5 = 5$*,* $F_4 = 3$*):*

```
theorem phi_fifth : goldenRatio^5 = 5 * goldenRatio + 3 := by
  have h2 : goldenRatio^2 = goldenRatio + 1 := gold_sq
  linear_combination
    (goldenRatio^3 + goldenRatio^2 + 2*goldenRatio + 3) * h2
```

3.3 Hammer 3: Transcendental Interval Analysis (π bounds)

Hammer 3 handles the floor identity $\lfloor 6\pi^5 \rfloor = 1836$. Mathlib provides tight rational bounds on π:

$$\texttt{pi_gt_d6} : 3.141592 < \pi, \tag{1}$$

$$\texttt{pi_lt_d4} : \pi < 3.1416. \tag{2}$$

Propagating through $\pi^2, \pi^4, \pi^5, 6\pi^5$ via `nlinarith` and `linarith` gives $1836 < 6\pi^5 < 1837$; the conclusion follows from `Int.floor_eq_iff`.

```
theorem floor_six_pi_fifth_eq_1836 : Int.floor (6 * Real.pi ^ 5) = 1836
    := by
  apply Int.floor_eq_iff.mpr
  constructor
  case left  => push_cast; linarith [six_pi_fifth_gt_1836]
  case right => push_cast; linarith [six_pi_fifth_lt_1837]
```

Lemma	Bound	Tactic
`pi_sq_lower`	$\pi^2 > 9.8696$	`nlinarith`
`pi_fourth_lower`	$\pi^4 > 97.409$	`mul_lt_mul`, `linarith`
`pi_fifth_lower`	$\pi^5 > 306.01$	`mul_lt_mul`, `linarith`
`pi_sq_upper`	$\pi^2 < 9.8701$	`nlinarith`
`pi_fourth_upper`	$\pi^4 < 97.42$	`mul_lt_mul`, `linarith`
`pi_fifth_upper`	$\pi^5 < 306.07$	`mul_lt_mul`, `linarith`

4 Coverage: Foundation Claims and Lean Files

Table 2 maps the BEDROCK and PILLAR foundation claims to their Lean 4 verification files.

Remark 3 (Eigenvalue convention)**.** *Two conventions coexist in the QHOTS literature.* Convention 1 *(Paper 1 [3]): define* $s_k^2 = N^2/(4\sin^2(2\pi k/N)) > 0$*, so that the elementary symmetric polynomials* $e_k(s^2)$ *are strictly positive and* $e_1 + e_2 \neq 0$. Convention 2 *(this paper, Lean library): work with the* signed *squared eigenvalues* $\lambda_k^2 < 0$ *of the antisymmetric part* A*, whose eigenvalues are purely imaginary; under this convention* $e_1(\lambda^2)+e_2(\lambda^2) = 0$*. The two are related by* $\lambda_k^2 = -s_k^2$*, which flips the sign of odd-degree elementary symmetric polynomials. PILLAR #70 is stated and verified under Convention 2.*

5 Key Results in Detail

5.1 The anticommutator $\{S, A\} = 0$ (BEDROCK #62)

The $\mathrm{GL}_2(\mathbb{Z}_{11})$ Qameah construction [4] produces an 11×11 integer matrix whose (i, j)-entry is

$$Q(i, j) = 11\big[(6i + 5j + 5) \bmod 11\big] + \big[6(i + j) \bmod 11\big] + 1.$$

Entries are a permutation of $\{1, \ldots, 121\}$ with magic constant $M_{11} = 671$. This $\mathrm{GL}_2(\mathbb{Z}_{11})$ construction is related to the standard Qameah square of the companion paper by a row/column permutation; the anticommutation $\{S, A\} = 0$ holds for both by the general circulant–Hankel theorem. (This equivalence is asserted but not formalized in the current Lean library; independent verification is available in the companion paper [3].) We define the centred symmetric and antisymmetric parts

$$2S(i, j) = Q(i, j) + Q(j, i) - 122, \qquad 2A(i, j) = Q(i, j) - Q(j, i).$$

Theorem 4 (BEDROCK #62)**.** *For all* $i, j \in \{0, \ldots, 10\}$*,*

$$\sum_{k=0}^{10} \big[2S(i, k) \cdot 2A(k, j) + 2A(i, k) \cdot 2S(k, j)\big] = 0.$$

The Lean proof is by `native_decide` over all 121 index pairs; the algebraic explanation (circulant–Hankel decomposition) is given in the companion paper [3].

Three supporting lemmas further characterize the Qameah structure: (i) $\sum_{i,j}(2S(i, j))^2 = 4840$ (*trace squared*), (ii) every row of $|2S|$ has sum 60, (iii) $121 \cdot S^2 + A^2 = 0$ entry-wise (BEDROCK #66).

5.2 The floor identity $\lfloor 6\pi^5 \rfloor = 1836$

The classical mass-ratio formula $\mu_0 = 6\pi^5 \approx 1836.118$ gives integer shadow $\lfloor 6\pi^5 \rfloor = 1836$. The proof constructs $1836 < 6\pi^5 < 1837$ from Mathlib's rational π bounds alone, without any floating-point oracle.

5.3 The Pochhammer generating function (PILLAR #71 / T95)

For the $N \times N$ Qameah magic square, define e_k as the k-th elementary symmetric polynomial of the squared eigenvalues $s_k^2 = N^2/(4\sin^2(2\pi k/N))$ of the symmetric part S). T95 states:

$$e_k = N^{4k} \cdot \frac{\binom{(N-1)/2+k}{2k}}{2k + 1}.$$

Verified integer instances:

[1]The Lean proof verifies $73205 + (-73205) = 0$ with signed eigenvalue convention. See Remark 3 below.

N	k	e_k	Proof
11	1	$11^4 \times 5 = 73205$	`norm_num`
11	2	$11^8 \times 7 = 1500512167$	`norm_num`
11	3	$11^{12} \times 4 = 12553713506884$	`norm_num`
11	4	$11^{16} \times 1 = 45949729863572161$	`norm_num`
7	1	$7^4 \times 2 = 4802$	`norm_num`
7	2	$7^8 \times 1 = 5764801$	`norm_num`
5	1	$5^4 \times 1 = 625$	`norm_num`

An additional ladder theorem verifies the 11-adic valuations $v_{11}(e_k) \geq 2k$ for $k = 1, \ldots, 4$, and the termination boundary $11 \nmid \binom{10}{10}$ is checked by `native_decide`.

5.4 The Clifford Bridge $S_3(11) = 2^{11}$ (PILLAR #55)

The sixth-power trigonometric sum $S_3(11) = 2048 = 2^{11}$ connects the Qameah order to Cl_{11}. Uniqueness—N=11 is the only odd prime $p \leq 47$ with $S_3(p) = 2^p$—is verified by `native_decide` over 14 primes.

5.5 D4–E8 pairing uniqueness (BEDROCK #67)

The equation $r^2 = 2h + 4$ selects exactly $\{D_4, E_8\}$ among all 27 simple Lie algebras up to rank 10 (by `decide`). A complementary theorem shows $2^{r-1} \cdot h - 3 = 3837$ uniquely selects $(r, h) = (8, 30)$.

6 Limitations and Future Work

6.1 Scope exclusions

Full G formula. Newton's constant is expressed in QHOTS as a product involving G_N, φ, and dimensional constants. Verifying agreement with the CODATA value at 1.3 ppm would require a formal real-number computation with approximately 7 significant digits of π, φ, and derived quantities—feasible in principle with tighter Mathlib π bounds and extended interval arithmetic, but not yet implemented.

Transcendental precision beyond 6 digits. Mathlib's current rational π bounds (`pi_gt_d6`, `pi_lt_d4`) provide 6 and 4 decimal digits respectively. The mass-ratio correction $\delta_{QED} \approx 1.7 \times 10^{-3}$ would require π to 10 or more digits to verify $\lfloor 6\pi^5(1+\delta_{QED}) \rfloor = 1836$ at the corrected level. The `Mathlib.Tactic.Polyrith` tactic and higher-precision π bounds may provide a path forward.

Spectral gap bounds. The claim that $\mathrm{spec}(L) \subset \{0\} \cup i\mathbb{R}$ for the $N \times N$ linear residue matrix L is proved symbolically in QHOTS (via the circulant–Hankel decomposition) but has not been formalized as a Lean statement, because the proof would require formalizing the theory of circulant matrices over $\mathbb{Z}/N\mathbb{Z}$.

Full spectral polynomial (characteristic polynomial of L). T91 states that the reduced characteristic polynomial of L for $N = 11$ is $x^{10} - 11^9$. The library verifies all numerical consequences of this (the values $e_1, \ldots, e_4$ and their divisibility) but does not verify the polynomial itself, which would require working with the charpoly API for matrices over $\mathbb{Z}$.

6.2 Paths to extension

- Using Mathlib's `Matrix.charpoly` to verify the characteristic polynomial $\det(\lambda I - L) = \lambda^{10} - 11^9$ directly.
- Using Lean's `Decidable` instances for modular arithmetic to extend the anticommutator proof to other odd primes N.
- Formalizing the generating-function identity T95 as a theorem about formal power series in $\mathbb{Q}[[x]]$ rather than a finite list of numerical instances.
- Connecting the D4–E8 pairing uniqueness to the McKay correspondence using Mathlib's developing representation theory.

7 Conclusion

The library—1,296 declarations, 76 files, zero `sorry`—covers the integer arithmetic skeleton of the QHOTS framework using three proof hammers: `norm_num`/`native_decide` (integer arithmetic), `linear_combination` (φ algebra), and `nlinarith`/`linarith` (π interval analysis). The main technical contribution is the machine-verified floor identity $\lfloor 6\pi^5 \rfloor = 1836$; complementary highlights are $\{S, A\} = 0$ for the 11×11 Qameah square and the Pochhammer formula T95 across three square orders. The library provides direct coverage for 20 Foundation claims while maintaining a clean separation between formalized mathematical identities and empirical comparisons with measured constants.

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Table 1: Named modules in the QHOTSLean library.

File	Thms	Principal content
`Basic.lean`	3	E8 / D4 constants as Lean definitions, $248 = 8 + 240$, magic constant $M_{11} = 671$
`Alpha.lean`	13	$\alpha = 28/3837$ arithmetic, master polynomial, Clifford Bridge $S_3(11) = 2^{11}$
`D4E8Pairing.lean`	2	$r^2 = 2h+4$ selects exactly $\{D_4, E_8\}$, 3837 uniqueness among D/E series
`GoldenMirror.lean`	6	$E_{\parallel} + E_{\perp} = 480$, ratios 2 and 3, complement symmetry
`GoldenMirrorAlgebraic.lean`	5	φ^n identities for $n = 1, \ldots, 5$ via `gold_sq`
`MuIntegerShadow.lean`	8	$\lfloor 6\pi^5 \rfloor = 1836$ via interval propagation
`SAAnticommutator.lean`	4	$\{S, A\} = 0$ for the 11×11 GL_2 Qameah square
`SALemmas.lean`	5	$\mathrm{tr}(S^2) = 1210$, $\lvert S \rvert$ row sums, $N^2S^2 + A^2 = 0$
`CrownJewels.lean`	5	G correction $27-7 = 20$, $\gcd(28\times1836, 3837) = 3$
`PochhammerIntegers.lean`	26	T95 Pochhammer formula, exact divisibility, 11-adic ladder
`Spectral.lean`	17	Spectral identities T91–T97, Vieta formula
`ProcaLagrangian.lean`	11	BEDROCK #69–71: $S_{YM} = 805255$, Proca $m^2 = 11$, $4L = S_{YM}$ residue
`ProcaHessian.lean`	19	Hessian integer arithmetic for Proca field
`MDS.lean`	27	Rank/nullity, $[[11, 1, 11]]$ MDS code, Singleton bound, nilpotency traces
`SpectralZeta.lean`	15	$\zeta_S(2n) = \binom{2n}{n}/N^{2n-1}$ integer skeleton
`EulerChar.lean`	8	$\chi(K_N) = N^2$ iff $N \in \{1, 11\}$, 2-complex counts
`LehmerQuintic.lean`	13	Minimal polynomial of $2\cos(2\pi/11)$, disc $= 11^4$, conductor
`NewtonSums.lean`	20	Odd $p_{2k-1} = -4^{k-1}$, even $p_{2k} = (11\binom{2k}{k} - 4^k)/2$
`NewtonIdentities.lean`	19	Newton's identities $k = 1, \ldots, 5$; Lehmer coeffs, Vieta consistency
`FourierDuality.lean`	33	$\lvert\hat{a}_p\rvert^2 = N^2\lvert\sigma_p\rvert^2$, trace chain $k = 1, \ldots, 4$
`CirculantHankel.lean`	23	Antisymmetric circulant anticommutes symmetric Hankel; $N = 11$ DFT skeleton
`Session67.lean`	64	$N = 11$ selection, S_{YM} cross-checks, $(5, 6, 11)$ number dictionary

Table 2: Foundation claims and Lean 4 coverage.

Claim ID	Statement	File
BEDROCK #58 / T91	$e_1 = 5 \times 11^4 = 73205$; $11^2 \mid e_k$	Spectral.lean
BEDROCK #60	$N^2 - 1 = 5! = 120$ (icosahedral, $N = 11$)	Tier3.lean
BEDROCK #61	$\varphi(30) = 8 = \mathrm{rank}(E_8)$	Tier3.lean
BEDROCK #62	$\{S, A\} = 0$ (11×11 GL_2 Qameah)	SAAnticommutator.lean
BEDROCK #63	$\mathrm{tr}(S^2) = 1210$ for $N = 11$	SALemmas.lean
BEDROCK #64	$\gcd(28, 3837) = 1$ (irreducibility)	Alpha.lean
BEDROCK #65	$128 \times 30 - 3 = 3837$ (spinor $\times$ Coxeter)	Alpha.lean
BEDROCK #66	$N^2 S^2 + A^2 = 0$ entry-wise, $N = 11$	SALemmas.lean
BEDROCK #67	$r^2 = 2h + 4 \Leftrightarrow (r, h) \in \{D_4, E_8\}$	D4E8Pairing.lean
BEDROCK #68	$\lvert S \rvert$ row sums $= (N^2 - 1)/4 = 30$	SALemmas.lean
PILLAR #55	$S_3(11) = 2048 = 2^{11}$ (Clifford Bridge)	Alpha.lean
PILLAR #57	$r = 8$ unique root of $r^3 = 4r + 480$	Alpha.lean
PILLAR #69 / T92	Vieta: $e_1 = N^4(N^2 - 1)/24$	Spectral.lean
PILLAR #70 / T93	$e_1(\lambda^2) + e_2(\lambda^2) = 0$ where $\lambda_k^2 < 0$ (signed; N=11)[1]	Spectral.lean
PILLAR #71 / T95	Pochhammer formula; $e_k = N^{4k}\binom{(N-1)/2+k}{2k}/(2k+1)$	PochhammerIntegers.lean
PILLAR #72 / T94	e_3 closed form for $N = 11$	Spectral.lean
PILLAR #73 / T97	Alternating N-adic valuation ladder	Spectral.lean
PILLAR #75	$E_\parallel + E_\perp = 480$ (E8 invariant sum)	GoldenMirror.lean
μ_0 shadow	$\lfloor 6\pi^5 \rfloor = 1836$	MuIntegerShadow.lean
Crown Jewel G	$27 - 7 = 20$, $\gcd(28 \times 1836, 3837) = 3$	CrownJewels.lean